APHL Pathogen Genomic Surveillance Capacity Assessment

A field-tested instrument for genomic surveillance capacity, built for reuse

What it is

A structured assessment system for laboratory capacity in pathogen genomic surveillance, built by APHL Global Health under the MADE Initiative. It has three parts:

- A **short form** sent to the laboratory two to three weeks ahead, completed by one senior respondent in about 30 minutes, used to scope the visit
- A **full baseline** administered on site over one to two days, covering infrastructure through financial sustainability, with application-specific modules that select automatically based on what the laboratory actually does
- An **institutional and national annex** administered once per site with leadership, covering the enabling environment

Both instruments run as standalone HTML files. No server, no license, no internet connection required. They work on a tablet at the bench and export structured data.

What it has produced

Three laboratories at the Research Institute for Tropical Medicine in Manila were assessed over three days in May 2026: the special pathogens laboratory, the national antimicrobial resistance reference laboratory, and the virology laboratory. Each received a full report. Findings reshaped the design of the training that followed.

The round also produced something less expected. It surfaced what the instrument itself could not yet capture, and that evidence is now driving a revision.

Five things that make it different

1. It routes every finding to the level that can act on it.

Roughly a third of what the first round surfaced was not a laboratory problem. Procurement lead times, employment modality for technical staff, data retention policy, multi-year funding, access and benefit sharing. Telling a laboratory to fix a national employment policy is not a useful recommendation.

Every finding now carries a constraint-level tag: actionable by the laboratory, by the institution, or above the institution. Reports are structured the same way, which means the national-level section can be quoted directly by laboratory leadership in upward communication. Partners have used it exactly that way.

For program design, this is the difference between an assessment that generates a training plan and one that also generates a policy dialogue agenda.

2. It measures continuity, not headcount.

Headcount tables are easy to collect but rarely informative. One laboratory in the round reported a fully staffed bioinformatics function with no vacancies. It had one bioinformatician, and the entire sequencing operation depended on that person.

The revised instrument asks, for each of twelve laboratory functions, how many people can perform it independently and what happens if the usual person is unavailable for two weeks. Single-point-of-failure exposure falls out of the answers rather than being asked about directly. It is also a more comfortable conversation than counting departures, which matters for the quality of what you get back.

For anyone thinking about sustainability and eventual transition, this is the metric that is usually missing.

3. It distinguishes what partners actually provide.

Most capacity instruments collect a partner list and a funding-source checkbox. That makes a laboratory whose entire portfolio consists of academic research collaborations look as diversified as one with three independent funders.

Each partnership is now recorded across several dimensions: what flows (cash to the laboratory, cash through government, reagents in kind, equipment, technical assistance, sequencing performed by the partner, network membership), through what channel, under what instrument, ending when, and what stops if it ends.

That yields four views directly usable for investment planning: funding concentration; the split between cash and in-kind and technical-only support; a calendar of agreements expiring within 18 months with no successor identified; and a map of single-point dependencies.

4. It is anchored to clinical laboratory standards, not to a general capacity checklist.

Questions are built against ISO 15189:2022 and the relevant CLSI guidance, including EP49 for clinical validity, MM24 for bacterial whole-genome sequencing and strain typing, and EP12-A2 for qualitative test performance.

One consequence worth naming. Most instruments ask whether assays are validated, which reliably returns yes. This one separates analytical validation from clinical validation, and captures validation status per assay rather than per laboratory. In the first round, every laboratory answered yes at the section level and reported that its metagenomic work was not validated for clinical use. Both statements were true. The instrument could not hold both, and now it can.

5. It produces two different outputs for two different audiences, without conflating them.

A narrative report goes to the laboratory and its partners. It is written to be useful to the laboratory, opens with strengths worth protecting, and separates findings by the level that can act.

A separate internal summary tracks domain-level maturity across a defined set of areas for program management and to measure change between assessment rounds. It is not a scorecard, does not carry a total score, and does not rank laboratories against each other.

Keeping these separate is deliberate. An instrument that produces a score a laboratory can be judged on changes what laboratories tell you.

Why it transfers

Nothing in the laboratory instrument is Philippines-specific. Application modules select based on the sequencing work a laboratory performs, so the same instrument serves a viral surveillance laboratory, a bacterial AMR reference laboratory, and a laboratory in its first year of build-out.

Country-specific policy context is captured in the institutional and national annex rather than in the laboratory instrument. That is the design decision that makes multi-country use possible: the laboratory instrument remains constant, making findings comparable across countries, while the annex accounts for the regulatory, procurement, and employment contexts that differ everywhere.

What a new country deployment would require, stated plainly:

- Assessor familiarization, roughly a half day, plus the assessor briefing document
- Two to three days on site for a three-laboratory site, one senior respondent per laboratory for the short form
- Access to institutional leadership for the annex, which is the piece that most affects output quality
- Local adaptation limited to the annex and to regulatory references

It does not require a data platform, a subscription, or a central server.

Where it is going

A revision is underway, driven by what the first round exposed. The substantive changes:

- A maturity scale replacing yes and no on a defined subset, so that in-draft, in-implementation and in-routine-use are distinguishable, and so that change between rounds is measurable
- Equipment recorded as owned, accessed elsewhere or non-functional, with runs per platform per year. The first round counted instruments and overstated capacity where a laboratory owned a sequencer it was not yet running
- Shared-resource mapping across co-located laboratories, which in the first round revealed that two laboratories named a third as their disaster recovery contingency, on the same campus and the same

power supply

- The function coverage, partner taxonomy and per-assay validation elements described above
- A reporting recipient matrix, which surfaced the most consistent operational gap in the round: sequencing results that are generated well and never reach the clinician, the national program, or the stewardship team

Target is a revised instrument set in the last quarter of 2026.

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